

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I Sunil Kumar I S (192324155) that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis

Given $p=11$, $q=13$

a. Compute n $\phi(n)$

$$n = p \times q = 11 \times 13 = 143$$

$$\phi(n) = (p-1)(q-1) = 10 \times 12 = 120$$

Answer:

- $n = 143$
- $\phi(n) = 120$

b. Choose $e=7$ and find the private key d

We need,

$$7d \equiv 1 \pmod{120}$$

$$7 \times 103 = 721$$

$$721 \bmod 120 = 1$$

Therefore,

$$\mathbf{d = 103}$$

$$\text{Private Key} = \mathbf{(103, 143)}$$

c. Encrypt the message $M=9$

Formula:

$$C = M^e \bmod n$$

$$C = 9^7 \bmod 143$$

$$9^7 = 4782969$$

$$4782969 \bmod 143 = \mathbf{48}$$

$$\mathbf{\text{Ciphertext} = 48}$$

d. Decrypt the ciphertext and verify the result

$$M = C^d \bmod n$$

$$M = 48^{103} \bmod 143 = \mathbf{9}$$

$$\text{Recovered Plaintext} = \mathbf{9}$$

e. Analyze how changing e affects security

- Smaller values of e provide faster encryption but may reduce security if used improperly.
- Larger values of e improve security but increase encryption time.
- The commonly used value is **65537** because it provides a good balance between security and efficiency.

2. Block Cipher Mode Evaluation

Given plaintext blocks:

$P=[1010,1010,1111,1010]$, Key = $K=1100$

a. Encrypt using ECB and CBC modes (Assume IV = 0000)

$P1 = 1010 \text{ XOR } 1100 = \mathbf{0110}$

$P2 = 1010 \text{ XOR } 1100 = \mathbf{0110}$

$P3 = 1111 \text{ XOR } 1100 = \mathbf{0011}$

$P4 = 1010 \text{ XOR } 1100 = \mathbf{0110}$

ECB Ciphertext:

0110, 0110, 0011, 0110

CBC Mode

Block 1

$1010 \text{ XOR } 0000 = 1010$

$1010 \text{ XOR } 1100 = \mathbf{0110}$

$C1 = 0110$

Block 2

$1010 \text{ XOR } 0110 = 1100$

$1100 \text{ XOR } 1100 = \mathbf{0000}$

$C2 = 0000$

Block 3

$1111 \text{ XOR } 0000 = 1111$

$1111 \text{ XOR } 1100 = \mathbf{0011}$

$C3 = 0011$

Block 4

$1010 \text{ XOR } 0011 = 1001$

$1001 \text{ XOR } 1100 = \mathbf{0101}$

C4 = 0101

CBC Ciphertext:

0110, 0000, 0011, 0101

b. Compare the ciphertext outputs

ECB Output:

0110, 0110, 0011, 0110

CBC Output:

0110, 0000, 0011, 0101

In ECB mode, identical plaintext blocks produce identical ciphertext blocks.

In CBC mode, identical plaintext blocks produce different ciphertext blocks because each block depends on the previous ciphertext.

c. Analyze which mode hides patterns better and explain why

CBC mode hides patterns better because every plaintext block is XORed with the previous ciphertext block before encryption.

ECB mode reveals repeated patterns and is therefore less secure.

3. DES vs 3-DES vs Blowfish Performance Study

Given:

- a. DES key size = 56 bits, time = 2 ms/block
- b. 3-DES key size = 168 bits, time = 6 ms/block
- c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

- i. Calculate total encryption time for each algorithm

DES

$$1000 \times 2 = \mathbf{2000\ ms = 2\ seconds}$$

3-DES

$$1000 \times 6 = \mathbf{6000 \text{ ms} = 6 \text{ seconds}}$$

Blowfish

$$1000 \times 1 = \mathbf{1000 \text{ ms} = 1 \text{ second}}$$

Algorithm	Total Time
DES	2 seconds
3-DES	6 seconds
Blowfish	1 second

ii. Analyze trade-offs between security and performance

- DES provides fast encryption but has weak security because of its small key size.
- 3-DES offers stronger security but has the slowest performance.
- Blowfish provides strong security with the fastest encryption speed among the three algorithms.

iii. Recommend the best algorithm for secure real-time communication

Blowfish is recommended for secure real-time communication because it provides strong security while requiring the least encryption time.

4. Diffie-Hellman Key Exchange Analysis

Given:

$$p=23, g=5$$

User A private key = 6

User B private key = 15

a. Compute public keys for A and B

Public Key of User A:

$$A = g^a \text{ mod } p$$

$$= 5^6 \text{ mod } 23$$

$$= \mathbf{8}$$

Public Key of A = **8**

Public Key of User B:

$$B = g^b \bmod p$$

$$= 5^{15} \bmod 23$$

$$= \mathbf{19}$$

Public Key of B = **19**

- b. Compute the shared secret key

User A:

$$K = 19^6 \bmod 23$$

$$= \mathbf{2}$$

User B:

$$K = 8^{15} \bmod 23$$

$$= \mathbf{2}$$

Shared Secret Key = 2

- c. Demonstrate how a Man-in-the-Middle attacker could alter the exchange

An attacker intercepts the exchanged public keys and replaces them with fake public keys. User A and User B unknowingly establish separate keys with the attacker. The attacker can read, modify, and forward all messages between them.

- d. Suggest a secure method to prevent the attack

The attack can be prevented using:

- Digital Certificates
- Digital Signatures
- Public Key Infrastructure (PKI)
- TLS/SSL Authentication

These methods verify the identity of users before exchanging keys.

5. ECC vs RSA Comparison

Given:

- a. RSA key size = 1024 bits
- b. ECC key size = 160 bits (equivalent security)
- c. Device processing capacity = limited (IoT device)
- d. Analyze computational cost for both systems

- RSA uses larger key sizes and requires more mathematical computations, making encryption and decryption slower.
- ECC provides the same security with much smaller key sizes, resulting in faster computation and lower processing requirements.

e. Estimate which consumes less power and memory

ECC requires:

- Smaller key sizes
- Less memory
- Lower CPU usage
- Lower battery power

RSA requires:

- Larger key sizes
- More memory
- Higher CPU usage
- More power consumption

f. Justify which cryptosystem is better for the given scenario with reasoning

ECC is the better choice for IoT devices because:

- It provides the same level of security as RSA with much smaller key sizes.
- It requires less processing power.
- It uses less memory.
- It consumes less battery power.
- It offers faster cryptographic operations on low-resource devices.

Conclusion:

ECC is the most suitable cryptosystem for IoT devices because it provides strong security with lower computational cost, lower memory usage, and reduced power consumption compared to RSA.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

strong
justification

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation